

GARIBALDI TILLAMOOK, OR, USA

WMO: 997706

Lat: 45.555N

Lon: 123.919W

Elev: 3

StdP: 101.29

Time Zone: -8.00 (NAP)

Period: 06-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-1.1	0.5	N/A	N/A	N/A	N/A	N/A	N/A	19.8	10.5	17.9	10.8	3.5	140	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.5	21.0	N/A	19.0	N/A	17.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	6.2	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.5	12.7	11.1	DB	-3.8	28.6	2.1	2.2	-5.3	30.2	-6.5	31.5	-7.7	32.7	-9.2	34.3	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.5	7.3	7.5	8.2	8.7	10.8	12.8	14.3	14.5	14.1	11.6	9.2	6.9		
	DBStd	3.58	2.68	2.70	2.14	2.15	2.05	1.61	1.33	1.31	2.03	2.14	2.71	3.19		
	HDD10.0	449	91	76	64	48	11	1	0	0	0	8	46	105		
	HDD18.3	2857	343	303	316	288	233	166	124	119	128	208	274	356		
	CDD10.0	636	6	6	7	10	37	84	134	140	123	58	21	8		
	CDD18.3	3	0	0	0	1	1	0	0	1	1	0	0	0		
	CDH23.3	30	0	0	0	4	9	0	4	3	11	0	0	0		
	CDH26.7	6	0	0	0	1	3	0	0	1	1	0	0	0		
(14) Wind	WSAvg	4.7	4.6	4.9	5.7	4.8	4.5	4.2	4.7	3.8	3.8	4.3	5.3	5.3		
(15) Precipitation	PrecAvg	2513	408	271	254	201	123	87	22	40	72	210	423	414		
	PrecMax	3352	694	638	535	387	194	159	69	150	249	375	911	734		
	PrecMin	1869	185	61	41	89	26	26	0	2	7	35	157	191		
	PrecStd	430	126	154	103	82	47	36	21	36	57	93	186	142		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.7	15.6	14.9	20.4	23.4	21.1	23.5	20.8	24.9	20.3	16.2	15.5		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	12.8	13.7	12.9	15.3	17.7	17.5	19.1	18.9	20.6	17.3	14.8	13.4		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	11.8	12.4	12.1	12.9	14.8	16.3	17.9	17.8	18.3	15.8	13.9	12.2		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	11.1	11.5	11.5	11.9	13.5	15.5	17.1	17.0	17.1	15.0	13.1	11.3		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.0	4.8	4.8	5.2	4.7	4.1	4.5	4.5	5.7	5.8	5.0	4.8		
		MCDBR	5.6	6.3	5.8	7.6	8.6	5.6	7.1	6.2	8.5	7.5	5.2	4.2		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.327	0.322	0.338	0.369	0.381	0.365	0.350	0.357	0.348	0.337	0.330	0.327		
	taud		2.505	2.546	2.514	2.391	2.381	2.447	2.514	2.497	2.523	2.549	2.517	2.478		
	Ebn at Noon		782	866	897	891	887	902	911	892	874	835	771	736		
	Edh at Noon		68	77	91	112	117	110	102	100	89	76	66	63		
(44) All-Sky Solar Radiation	RadAvg		1.19	2.05	3.04	4.38	5.42	5.91	6.36	5.46	4.27	2.55	1.40	0.99		
	RadStd		0.17	0.26	0.35	0.37	0.48	0.48	0.50	0.41	0.35	0.29	0.18	0.14		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (26 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+22

Nomenclature: See separate page